NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

Assignment:- 03

Submitted to:

Ms. Ayesha Zafar

Submitted by:

Zulqarnain Hassan

Roll No: …………………… SP21379

Subject: …………………… DBMS

Department: ……………… BSSE-4-Afternoon

Date: ……………………… 25-04-2024

Database normalization:

Database normalization is a process that optimize database ~~efficiently~~ and prevent structure by reducing data redundancy and improving data integrity.

Normalization is a set of rules and guidelines that help organize data efficiently and prevent common data anomalies like update anomalies, insertion anomalies, and deletion anomalies.

There are several levels, referred to as normal form. The most commonly used normal form are:

(i) First Normal Form (1NF)
(ii) Second Normal Form (2NF)
(iii) Third Normal Form (3NF)
(iv) Fourth Normal Form (4NF)
(v) Fifth Normal Form (5NF)

First Normal Form (1NF)

1NF ensures that each column in a table contains atomic, indivisible values. There should be no repeating groups, and each column should have a unique name.

unformalized Form:
Consider an unnormalized table that stores customer orders.

ID	customer	address	date	Product
1	John	XYZ	14/01	Apples, oranges
2	John	XYZ	15/01	Graps, strawberries
3	Alice	XYZ	18/01	Lemons, Limes
4	Bob	XYZ	19/02	cloths, shirts
5	Bob	XYZ	3/03	Pens,

The products column contains a list of items, To bring it to 1NF, we split the products into separate rows:

ID	customer	address	date	Product
1	John	XYZ	14/01	Apples
1	John	XYZ	14/01	oranges
2	John	XYZ	15/01	Graps
2	John	XYZ	15/01	strowberries
3	Alice	XYZ	18/01	Lemons
3	Alice	XYZ	18/01	Limes
4	Bob	XYZ	19/02	cloths
4	Bob	XYZ	19/02	shirts
5	Bob	XYZ	3/03	Pens.

Second normalise Form:

Building on 1NF, 2NF eliminates Partial dependencies. A table is in 2NF if it's in 2NF and all non-Key attributes are functionally dependent on the primery Key, but not on other non-key attributes.

SID	Sname	CID	cname	instructor
1	John	101	math	P. smith
2	Alice	102	Physics	P. Johnson
3	Bob	104	History	P. smith
4	Ben	103	DBMS	P. Davis
5	Harry	105	SDA	P. Naveed

This table violates 2NF because the instructor attribute depends on both SID and CID. To achieve 2NF, we split the table into two separate tables.

student table:

SID	SName
1	john
2	Alice
3	Bob
4	Ben
5	Harry

Course table:

CID	cname	instructor
101	math	P. smith
102	Physics	P. johnson
104	History	P. smith
103	DBMS	P. Davis
105	SDA	P. Naveed

Third Normal Form:

It eliminates transitive dependencies. All non-key attributes are functionally dependent on the primery key, but not on other non-key attributes.

EID	Ename	PID	Pname	manager
1	John	101	A	jak
2	Alice	102	B	jak
3	Bob	103	C	jak
4	Ben	104	D	jak
5	Harry	105	E	jak

This table violates 3NF because the Ename attribute depends on the EID, not directly on the primary key. To bring it to 3NF, we split the table into two separate tables -

Employees table:

EID	Ename
1	John
2	Alice
3	Bob
4	Ben
5	Harry

EmplyeeProject table:

EID	PID
1	101
2	102
3	103
4	104
5	105

Projects table

PID	P Name
101	A
102	B
103	C
104	D
105	E

Forth Normal Form:

Deals with multi valued dependencies. where an attribute depends on another attribute but not is not a Function of the primary Key.

PID	customer	CID	date	Product
1	John	101	14/01	Apple, oranges
2	Alice	102	2/06	Graps, strawberries
3	joy	103	3/06	Lemon, ~~[illegible]~~
4	Ben	104	18/07	~~[illegible]~~ Apple
5	Harry	105	d/08	oranges, Graps

There is a multi-valued dependency b/w PID and Product. To achieve 4NF, we split the table into three separate tables:

Customer ~~[illegible]~~ Table

PID	customer
1	John
2	Alice
3	Joy
4	Ben
5	Harry

Product table

PID	product
1	Apple
2	orange
3	Graps
4	Lemon
5	strawberries

CustomerProduct table

PID	CID
1	101
2	102
3	103
4	104
5	105

Fifth Normal Form:

These forms deal with cases where a table is in 4NF, but there are join dependencies that can be Further optimized.

A table is in fifth normal form if:

- It is in its forurth normal from
- It cannot be subdivided into any smaller tables without losing some form of information.